
Topological comparison of social networks: modularity in institutional emails and on Facebook

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Abstract

This research comparatively analyzes an institutional email network and a Facebook friendship network, based on the hypothesis that the former, due to its formal hierarchical structure, would exhibit more clearly defined communities. Both networks were topologically characterized, compared with classical generative models, centrality measures were computed, robustness under different types of attacks was evaluated, and community detection was applied using the Louvain algorithm. In the email network, the correspondence between communities and departments was also examined, as well as class assortativity. The results show that, although the institutional network clearly reflects the organizational structure, it is the Facebook network that exhibits greater modularity and more clearly defined clusters, which contradicts the initial hypothesis.

1 Introduction

Complex social networks are present in multiple areas of everyday life, from personal relationships to the organizational structures of institutions. Understanding how these networks are organized and, in particular, how communities form within them, is fundamental for the analysis of phenomena such as information diffusion, collaboration, and internal segmentation. From the perspective of Graph Theory Barabási [2016], the concept of a community refers to a subset of nodes that exhibit a higher density of connections among themselves than with the rest of the network. In topological terms, it can be interpreted as a subgraph with a high density of internal links.

In this work, community formation is analyzed in two complex social networks with different structural characteristics. On the one hand, an email network in a European institution is considered, characterized as undirected, unweighted, and disconnected, built from message exchanges among members of that organization. Each node represents a person and each edge indicates at least one exchanged email. In addition, labels are available identifying the institutional affiliation of each node. On the other hand, a Facebook network is examined, also undirected and unweighted, but connected, built from data collected through a survey within the platform. In this case, each node represents a user profile and edges reflect friendship relationships between peers.

Previous studies indicate that natural social networks, such as Facebook, tend to exhibit spontaneously formed communities, with fuzzy boundaries and overlapping structures Girvan and Newman [2002]. In contrast, networks arising in formal or institutional contexts usually show more clearly defined community divisions, reflecting underlying organizational structures. In email networks it has been shown that formal hierarchy influences interaction patterns, generating communication dynamics structured by hierarchical levels Sims et al. [2014], which can lead to more pronounced community partitions. In the university setting, communication between previously defined organizational units has been observed to follow a superlinear structure, indicative of high modularity Mathiesen et al.

[2013]. Conversely, networks such as Facebook are structured around personal affinities and multiple simultaneous memberships, without reflecting clear hierarchical structures Bedi and Sharma [2016].

The objective of this study is to evaluate whether community formation is clearer in the institutional email network than in the Facebook network. The hypothesis is that the institutional network, being associated with a formal and hierarchical organization, will exhibit higher modularity. To test this hypothesis, a topological characterization of both networks will be performed, they will be compared with classical generative models (Erdős-Rényi, Watts-Strogatz, and Barabási-Albert), different centrality measures will be analyzed, their structural robustness to node removal will be evaluated, and Louvain community detection algorithms will be applied. Likewise, a comparison between the detected communities and the institutional departments in the email network will be carried out, complemented with an evaluation of class assortativity in that network. These tools will allow an empirical test of the proposed hypothesis and a discussion of the relationship between organizational structure and community cohesion in different types of social networks.

2 Methods

Two datasets corresponding to complex social networks were used: an email network from a European institution and a Facebook friendship network. In the case of the email network, since it is not connected, its giant component was extracted to enable structural analysis.

First, the basic properties of both networks were verified, such as number of nodes, number of edges, density, average degree, clustering coefficient, average shortest path length, and diameter. These metrics allowed an initial topological characterization of the observed structures.

Subsequently, synthetic networks were generated using the classical Erdős-Rényi, Barabási-Albert, and Watts-Strogatz models, with parameters equivalent to the real networks, in order to contrast the observed empirical properties with theoretical configurations. The comparison focused on average degree, mean clustering, and average shortest path length.

To deepen the topological analysis, different centrality measures were computed for both networks: degree centrality, betweenness, closeness, and eigenvector centrality. These measures were graphically represented with visual encoding based on node size and color, allowing a simultaneous interpretation of multiple nodal attributes.

The structural robustness of both networks under node removal was also analyzed, considering three types of attacks: random, degree-centrality targeted, and betweenness-centrality targeted. For each type of attack, the reduction in the size of the giant component and the drop in global network efficiency were evaluated.

Community analysis was performed using the Louvain algorithm, which partitions the graph by maximizing modularity. The modularity obtained in each network was evaluated, and the detected communities in the institutional network were compared with the known departments of each node, using a Chi-square test of independence and Cramér's V coefficient to quantify the degree of association between both categorical variables.

Finally, class assortativity was computed for the email network, with the objective of evaluating whether nodes tend to connect with others that belong to the same department, allowing interpretation of the degree of intra- and inter-departmental segmentation in the analyzed organization.

3 Results and Discussion

3.1 Description of the selected datasets

For the development of this work, the following datasets were selected:

- **Facebook friendship network based on a survey**
- **Email network of a European research institution**

The **Facebook friendship network (Facebook Network)** is an undirected and unweighted network of contact circles. The data were collected through a survey conducted in the application. Each node is a profile and an edge represents a friendship relationship. It contains 4039 nodes and 88234 edges.

The **email network of a European research institution (Email Network)** is an undirected and unweighted network of contacts via email in that organization. Each node represents a person and an edge means that at least one email was exchanged. In this dataset, the department to which each node belongs within the institution is known. The general characteristics of the networks can be found in Table 1

Table 1: Topological variables of the networks

Metric	Facebook	Email-EU
Number of nodes	4039	1005
Number of edges	88234	16706
Directed	No	No
Weighted	No	No
Connected	Yes	No
Density	0.0108	0.0331
Average degree	43.6910	32.58
Average clustering coefficient	0.6055	0.4071
Average shortest path length	3.705	2.571
Diameter	8	7

The Email network is disconnected; therefore, in what follows we will use its giant component to perform the analyses. This network has higher density than the Facebook network, which implies a larger number of interactions among people. The average shortest path length is smaller in the Email network, which is consistent with a formal organization in which communication among its members is more efficient.

The Facebook network has a higher clustering coefficient and average degree, which suggests, in principle, the existence of local community structures. This justifies the use of more appropriate methods to address the proposed hypothesis about community formation and the corresponding modularity in each network.

3.2 Comparison with prototypes

Analyzing the comparison between real networks and theoretical models is fundamental to understand whether the observed structures can be explained by simple mechanisms or whether they reflect more complex patterns, such as those that could arise in organizational or natural social contexts. This type of analysis makes it possible to empirically place each network within a broader structural framework, which is key to evaluating the validity and scope of the proposed hypothesis.

With the aim of interpreting the structure of the real Facebook and Email networks, a comparison was carried out with three classical theoretical network models: *Erdős-Rényi* (ER), *Barabási-Albert* (BA), and *Watts-Strogatz* (WS). Three key metrics were evaluated: *average degree*, *average clustering coefficient*, and *average shortest path length*. These allow analysis of the global and local behavior of networks in terms of connectivity and structural cohesion.

In the case of Facebook, the average degree of the real network (Table 2) was well replicated by the ER and BA models, which yielded similar values. However, the average clustering of the real network was much higher than that predicted by these models. This discrepancy suggests that

Facebook exhibits a "small-world" type structure, characterized by a high local density of connections combined with short paths between nodes. The ER and BA models, which assume pure randomness or growth by preferential attachment, respectively, fail to capture this essential property. In contrast, the Watts–Strogatz model, designed to reproduce "small-world" topologies, matches much better in both degree and clustering, making it a more suitable representation of the Facebook network in terms of local structure.

In the Email network, the ER and BA models (Table 2) again fit the average degree well, but strongly underestimated the average clustering. This difference suggests that the Email network also exhibits non-random structural patterns, possibly linked to functional or organizational segmentations, such as division by departments. The Watts–Strogatz model managed to replicate both the average degree and a high clustering (similar to the real one), but the latter could overestimate local cohesion if the Email network structure is more influenced by formal hierarchies than by spontaneous triangles. In this case, the Watts–Strogatz model cannot generate modularity based on institutional divisions, but only on local density.

In summary, although the ER and BA models approximate the average degree of the real networks reasonably well, they fail to reproduce the observed clustering, highlighting their limitation in capturing local structure and community formation. The Watts–Strogatz model provides a better representation of clustering in both networks, especially in Facebook, but may overestimate local cohesion in the case of more hierarchical institutional structures such as the Email network. These findings support the use of complementary methods such as community detection and centrality analysis to capture the structural complexity of real social networks beyond what simple generative models can offer.

Table 2: Comparison with theoretical models (ER, BA, WS) and real networks

Model	Metric	Facebook	Email-EU
Real network	Average degree	43.69	32.58
	Average clustering	0.6055	0.4071
	Average shortest path length	3.705	2.571
Erdős–Rényi	Average degree	43.64	31.97
	Average clustering	0.0108	0.0317
	Average shortest path length	2.6076	2.3084
Barabási–Albert	Average degree	41.78	29.55
	Average clustering	0.0372	0.0774
	Average shortest path length	2.5338	2.3232
Watts–Strogatz	Average degree	42.00	30.00
	Average clustering	0.5363	0.5323
	Average shortest path length	2.9967	2.7967

The topological metrics of the real networks are compared with those of networks generated by three classical models. The average degree, average clustering coefficient, and average shortest path length are reported.

3.3 Node centrality analysis

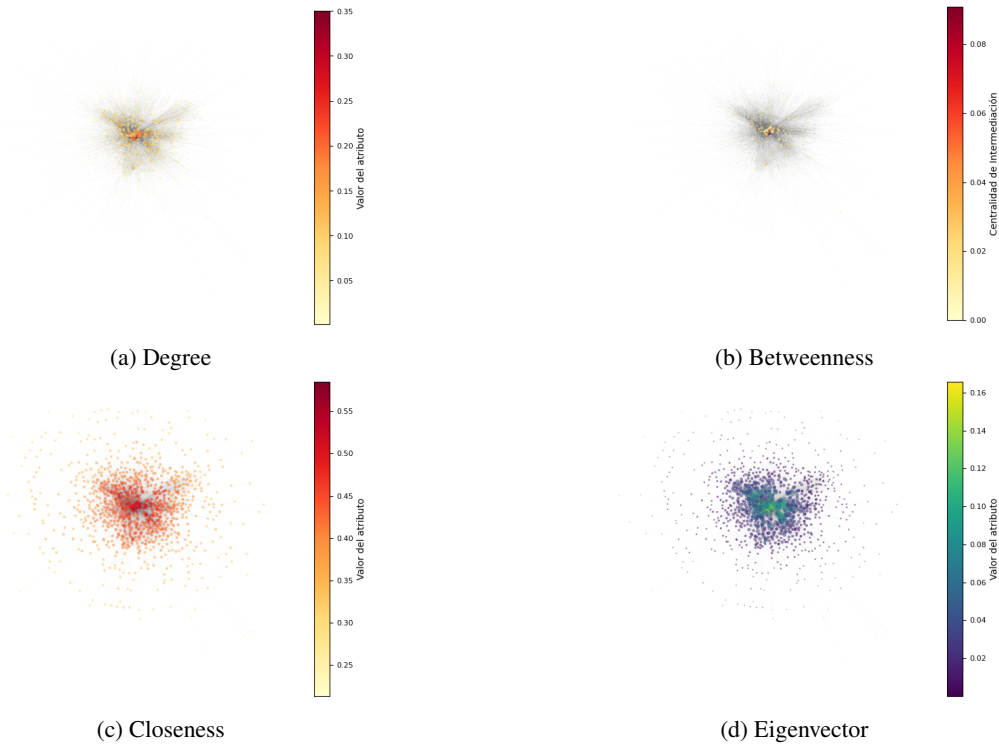


Figure 1: Centrality measures in the Email network.

Different centrality measures were analyzed on the Email and Facebook networks in order to characterize their internal topological structure.

First, degree centrality analysis in the Email network reveals a dense core with highly connected nodes to the rest of the network (see Figure 1). This pattern suggests the existence of an organizational center that concentrates a large portion of interactions. Toward the periphery, nodes with low connectivity are observed, which is typical of hierarchical structures where a few actors concentrate most ties. In contrast, in the Facebook network (see Figure 2) multiple high-degree nodes are identified, distributed across different communities. This modular configuration indicates a more decentralized distribution of connectivity, with several “local centers” of activity.

Regarding betweenness centrality, the Email network again presents a compact core with several nodes of high betweenness, surrounded by many peripheral nodes with low values (see Figure 1). This pattern is consistent with a segmented institutional structure, in which certain individuals act as key intermediaries in communication between departments. In contrast, in the Facebook network (see Figure 2), nodes with high betweenness are located mainly between dense communities. They are not necessarily the most connected, but they play a strategic role by connecting different groups, functioning as structural bridges.

In relation to closeness centrality, the Email network (see Figure 1) shows a centralized topology: most nodes present high closeness centrality and are located in the core of the network, from where the rest of the network can be reached quickly. This is consistent with a network that seeks to maximize communication efficiency. Peripheral nodes show lower closeness, in line with hierarchical or radially organized networks. In the Facebook network (see Figure 2), on the other hand, nodes with higher closeness tend to be located at the center of specific communities or in positions that connect modules. This suggests distributed communicational efficiency.

Finally, eigenvector centrality was analyzed in the Email network (see Figure 1) and it was found that the most influential nodes concentrate in the central core, being highly interconnected with each other. This also reflects a hierarchical structure where a few people occupy more relevant roles

in communications. On the other hand, in the Facebook network (see Figure 2), high eigenvector-centrality nodes are distributed among communities, each with its own group of relevant nodes. This is consistent with a decentralized and modular network, where there is no single dominant core, but rather multiple centers of relative importance.

The analyzed centrality measures are complementary, since each captures different aspects of network structure. Degree centrality identifies nodes with high local activity, betweenness highlights those that connect communities, closeness evaluates communication efficiency from each node, and eigenvector centrality highlights influential nodes by being connected to other important ones. Their combined use enables a more complete characterization of the analyzed networks.

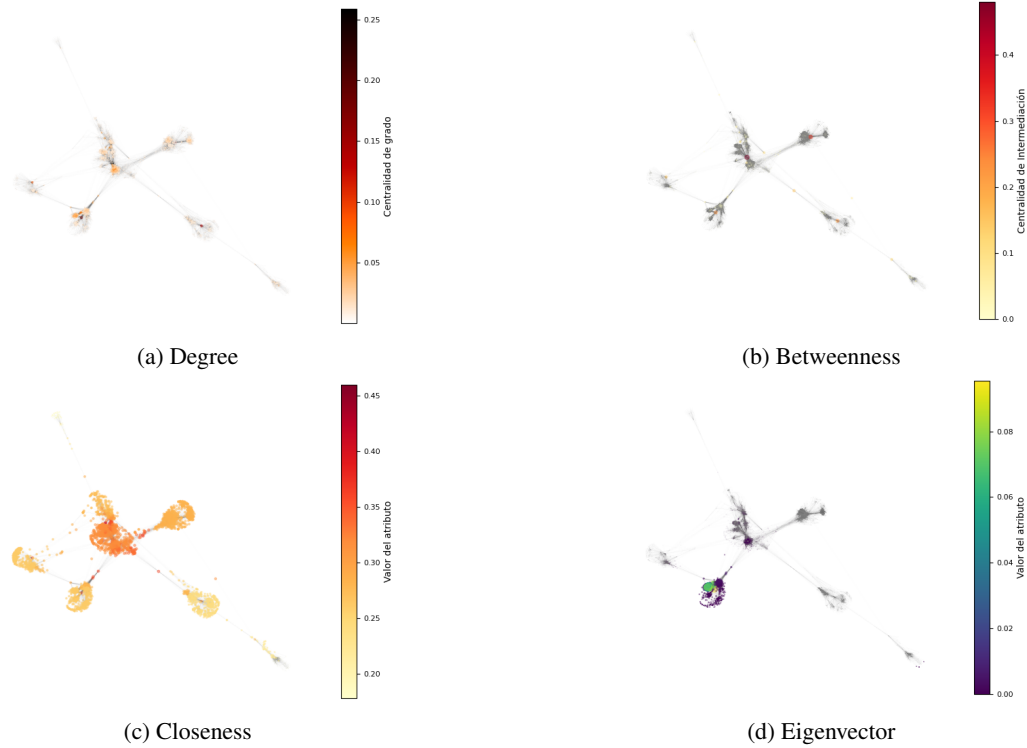


Figure 2: Centrality measures in the Facebook network.

3.4 Graph robustness analysis

The **Graph Robustness Analysis** is used to study how vulnerable or resistant a network is to the removal of nodes or edges. This type of analysis aims to understand how networks behave under attacks or disconnections.

In **Figure 3** the results of the robustness analysis for the Facebook network and the Email network are shown, considering random attacks, attacks targeted at nodes with the highest degree centrality, and attacks targeted at nodes with the highest betweenness centrality. In **Figure 3 a)** the relative size of the giant component over the total number of nodes is shown as a function of the fraction of removed nodes. In **Figure 3 b)** global efficiency is shown as a function of the fraction of removed nodes.

Robustez rápida: Email-EU

Componente gigante

Y-axis: N_g / N

X-axis: Fracción nodos procesados

Legend: aleatorio (blue), grado (orange), betweenness (green)

Eficiencia global (apx)

Y-axis: Eficiencia global (apx)

X-axis: Fracción nodos procesados

Legend: aleatorio (blue), grado (orange), betweenness (green)

Robustez lenta: Email-EU

Componente gigante

Y-axis: N_g / N

X-axis: Fracción nodos procesados

Legend: aleatorio (blue), grado (orange), betweenness (green)

Eficiencia global (apx)

Y-axis: Eficiencia global (apx)

X-axis: Fracción nodos procesados

Legend: aleatorio (blue), grado (orange), betweenness (green)

Figure 3: Robustness analysis | Targeted attacks by Degree centrality, Betweenness, and Random attacks.

Through the robustness analysis it is possible to visualize how, for all types of attacks, the relative size of the giant component and the global efficiency decrease as more nodes are removed from the network.

In the Facebook network, the most aggressive attack is the one targeted at nodes with the highest betweenness centrality, because the nodes through which many shortest paths pass are attacked first. When such nodes are removed, the network fragments into smaller components and the network size (or its giant component) tends to decrease more. Regarding the Email network, although centrality-targeted attacks are more aggressive than random attacks, the impact of the targeted attack by degree or betweenness centrality is similar. On the other hand, a critical point is observed in the fraction of removed nodes where the relative size of the giant component drops abruptly.

Regarding global efficiency, there is a similar trend to that found when analyzing the relative size of the network (or its corresponding giant component) as a function of the number of removed nodes.

In all cases, the random attack is the least effective, which is expected in "small-world" type networks.

3.5 Communities in graphs

We proposed to analyze the community structure present in both networks through the detection of node groupings with high density of internal connections and low external connectivity. This type of partition allows the identification of latent structures that often reflect social, organizational, or functional relationships.

To this end, the Louvain algorithm was applied, recognized for its efficiency and its ability to maximize the modularity of the partition, a measure that evaluates how well defined the boundaries between communities are.

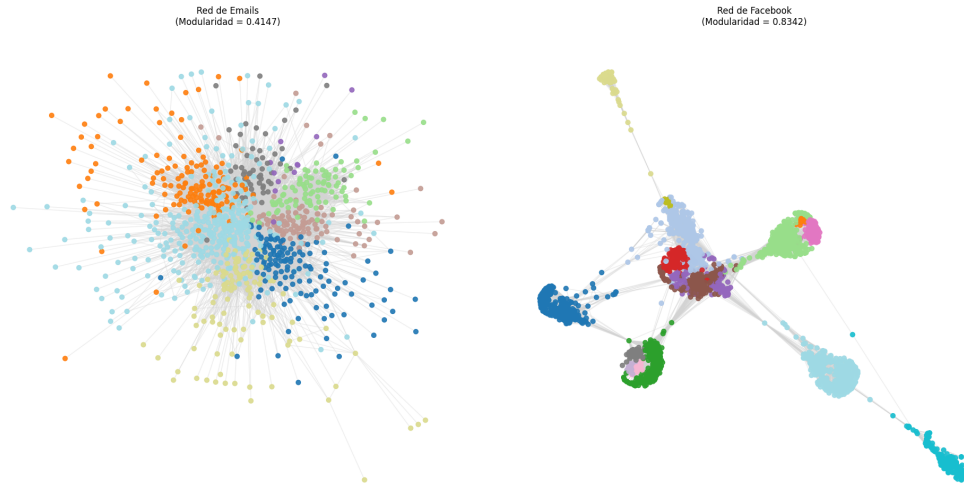


Figure 4: Communities found by the Louvain algorithm

When comparing the results (Figure 4), a significant difference in the community organization of both networks becomes evident. The Facebook network presents a highly modular structure ($Q = 0.8342$), with well-defined communities and a marked internal hierarchy, which is characteristic of social networks based on affinity ties. By contrast, the Email network shows moderate modularity ($Q = 0.4147$), with more diffuse communities and lower internal segmentation. This pattern corresponds to functional environments where communication between different sectors is common and necessary.

While in the Facebook network several communities reveal significant substructures (internal modularity > 0.5), in the Email network only one community presented notable substructure. These results suggest that the social network tends to organize into more clearly defined clusters, whereas the email network reflects a more integrated and transversal organization.

3.5.1 Relationship between Communities and Departments in the Email Network

To determine whether there is a relationship between the detected communities and the departments to which employees belong within the institution, a hypothesis test of independence between these variables is proposed. After building the contingency table with the cross frequencies of nodes by community and department, the Chi-square statistical test was applied. Cramér's V coefficient was also computed, a statistical measure that indicates the strength of association between two categorical variables. It is based on the chi-square statistic and provides a normalized value between 0 and 1, where 0 indicates no association and 1 a perfect association.

The obtained results were the following:

- Chi-square: 5535
- p-value < 0.001
- Cramér's V: 0.896 (> 0.6 : indicates strong association)

On the other hand, when analyzing the distribution of nodes from each department across communities (Figure 5), it was found that the communities are formed by people from specific departments. Likewise, it is notable that most members of a given department are always in a single community. These observations, together with the low p-value obtained in the independence test and the Cramér's V coefficient close to 1, allow us to conclude with considerable confidence that the community structure of the network is strongly linked to the formal organization of the institution it represents.

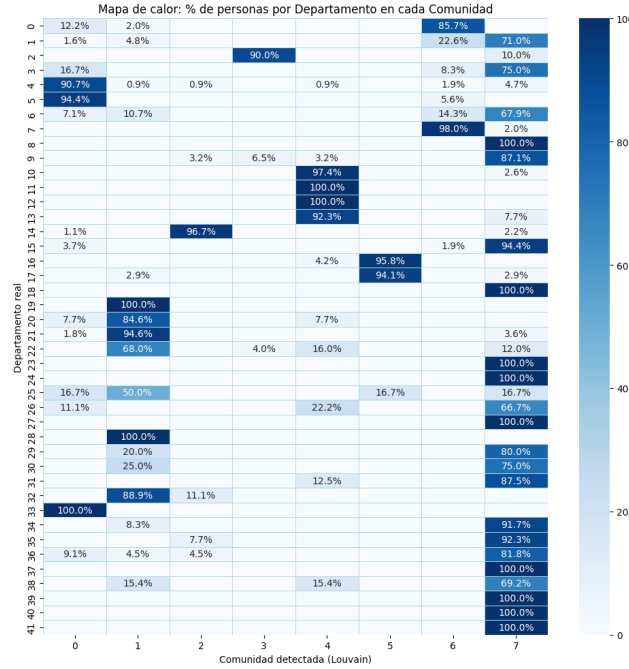


Figure 5: Heatmap: distribution of Departments [% of people] by Community.

3.5.2 Class assortativity of the Email Network

Class assortativity in a network is a value that indicates the tendency of nodes to connect with similar or different nodes in terms of a specific characteristic. A value close to +1 suggests that nodes tend to connect with others that share the same characteristic (assortativity), while a value close to -1 indicates that nodes tend to connect with nodes that have different characteristics (disassortativity). A value close to 0 suggests a random network, where node connections are not determined by similarity of characteristics.

Based on the department to which each node belongs, class assortativity was computed for the Email network, obtaining a value of **0.3024**. This value close to 0 indicates that there is no strong tendency for members of departments to communicate exclusively with people within the same department nor, conversely, to communicate only with those outside their department.

On the other hand, although not very pronounced, the obtained value indicates a greater degree of intra-departmental communication than communication between different departments. This finding is consistent with the results obtained when comparing departments with the detected communities, where the presence of entire departments within communities was found.

4 Conclusions

When comparing the studied networks with the Erdős-Rényi, Barabási-Albert, and Watts-Strogatz models, they fitted best to the latter, which reproduces "small-world" properties with high local clustering, indicating non-random structures.

The Facebook network shows a more decentralized and modular structure, with multiple local centers of activity and nodes that act as bridges between communities. It has a higher clustering coefficient, suggesting more naturally cohesive local communities.

The Email network presents a centralized structure with a dense core of highly connected nodes, which is typical of hierarchical organizations. Although it has a lower clustering coefficient than Facebook, a strong correlation was found between the detected communities and the formal departments of the institution. On the other hand, the positive class assortativity indicates a slight preference for intra-departmental communication.

Regarding robustness analysis, the Facebook network was more vulnerable to attacks targeted at nodes with high betweenness, while the email network showed similar vulnerability to degree- and betweenness-based attacks. Both were resilient to random failures.

Community analysis revealed key differences between both networks. While the social network showed an organization into well-defined groups and a marked internal hierarchy, the email network presented a more integrated and transversal structure. This suggests that personal ties tend to form closed clusters, whereas functional networks require interconnection between different sectors.

The results obtained in this analysis point to rejecting the initial hypothesis that community formation is clearer in the email network than in the Facebook network, and that it should exhibit greater modularity. Despite this, it can be concluded that the Email network, based on institutional interactions, clearly reflects the underlying organizational structure, showing modularity linked to formal divisions, which contrasts with the Facebook network, where communities emerge in a more spontaneous and decentralized manner.

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